

VANGUARD AEROSPACE: RMD-01 POWER & PROPULSION INTERFACE

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DIRECTOR'S NOTE: *This document contains the most current specifications for the TARTARUS-1 power and propulsion systems. All personnel with access must acknowledge the 2026-03-14 revision prior to any reactor interaction. The changes to Section 4 (Casimir Cavity Fluctuation Matrix) reflect the incident of 2026-02-27. — Dr. Vasquez*

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1. CORE POWER ARCHITECTURE

The recovered NHI craft (Designation: **TARTARUS-1**) does not utilize chemical combustion or nuclear fission in any conventional sense. Propulsion is derived from a **Zero-Point Energy (ZPE) Casimir Cavity Array** catalyzed by the alpha-decay of synthetic **Isotope 115 (Moscovium-299)**. The theoretical framework underpinning this

system is the **Volkov-Zel'dovich Exotic Vacuum Model** (classified), which posits that Element 115 acts as a catalyst for vacuum fluctuation amplification, converting virtual particles into usable graviton flux.

The ZPE harvesting mechanism operates on principles inconsistent with the Standard Model of particle physics. Per RMD-02 §1, the spatial manifold manipulation requires a minimum graviton flux of 4.72 TeV to sustain even a Class I fold. The power architecture is therefore the critical bottleneck for all downstream systems, including navigation (RMD-02), neural synchronization (RMD-03), and atmospheric maneuvering (RMD-04).

1.1 Element 115 Decay Chamber

The reactor core requires a steady feed of Moscovium-299 (Mc-299), a synthetic superheavy element with a half-life of 2.6 seconds under standard conditions. Within the decay chamber, Mc-299 is stabilized via the **Strong Nuclear Confinement Lattice (SNCL)**, a crystalline matrix of neutron-degenerate matter that extends the effective half-life to approximately 8.3 hours. The decay process generates a localized gravity wave which is harvested by the Casimir Array.

Parameter	Nominal Value	Warning Threshold	Critical Threshold	Failure Consequence
Core Temperature	4.2 K	> 10 K	> 15 K	Superconductor quench, ZPE collapse, total power loss
Decay Rate	4.72 TeV	< 4.60 TeV	< 4.50 TeV	Insufficient graviton flux for spatial folding (RMD-02 §1)
Decay Rate (Upper)	4.72 TeV	> 4.90 TeV	> 5.00 TeV	Localized false-vacuum decay (Event Horizon formation)
Moscovium Feed	12.5 mg/hr	< 11.0 mg/hr	< 10.0 mg/hr	Gravity wave inversion, hull implosion within 8.2 seconds
SNCL Lattice Integrity	99.7%	< 98.0%	< 95.0%	Uncontrolled alpha-decay cascade, radiation breach
Casimir Cavity Gap	10.0 nm ± 0.02	± 0.05 nm	± 0.10 nm	Vacuum fluctuation phase decoherence, power loss
Magnetic Confinement Field (MCF)	98.5%	< 94.0%	< 50.0%	Element 115 runaway fission → black hole event (RMD-05 §2)

DIRECTOR’S NOTE: *The MCF threshold of 50% was revised downward from 60% following the Monte Carlo simulations of 2026-01-19. The original threshold produced excessive false positives. — Dr. Vasquez*

1.2 Isotope 115 Decay Chain Byproducts

The alpha-decay of Mc-299 produces a cascade of daughter isotopes, each with distinct radiological and exotic properties. The following table details the primary decay chain and required containment protocols:

Daughter Isotope	Half-Life	Radiation Type	Energy (MeV)	Exotic Property	Containment Protocol	RMD Reference
Mc-295	0.59 s	α	8.91	Graviton Emission	SNCL immediate capture	RMD-01 §1.1
Mc-291	1.2 s	$\alpha + \beta^-$	7.34	Temporal Shear Field	Liquid Helium bath at 2.1 K	RMD-05 §1
Bh-291	3.8 s	$\alpha + \gamma$	6.88	Strong Force Resonance	Lead-Bismuth shielding (12 cm)	N/A
Db-287	8.4 s	β^-	4.22	Neutrino Burst	Bioelectric dampening field	RMD-03 §2
Nb-283	14.6 s	$\beta^- + \gamma$	3.15	Secondary Tachyon Emission	Tachyon siphon to QEA buffer	RMD-02 §1
Y-279	22.0 s	$\beta^- + \gamma$	2.87	Standard radiological	Standard lead containment	N/A
Sr-275	48.3 s	$\beta^- + \alpha$	1.93	Alpha emission (secondary)	Double-walled vacuum vessel	N/A

Critical Note: If the SNCL fails (§1.1), the full decay chain completes in under 48 seconds, producing a cumulative tachyon flux of **14.7 micro-Sieverts** — sufficient to cause localized temporal displacement per RMD-05 §1.1. Personnel must initiate emergency evacuation per RMD-05 §3 (Protocol Omega) if tachyon flux exceeds 8.0 micro-Sieverts at any point within the containment perimeter.

1.3 Gravity Wave Amplifiers (GWA)

The GWA translates the ZPE output into directional thrust by projecting graviton wavefronts through a phased array of **Kerr-Newman Metric Projectors**. The GWA is capable of producing thrust vectors in any direction without rotating the craft, as the graviton field decouples the thrust vector from the hull orientation.

Pilots must never engage the GWA while the **Magnetic Confinement Field (MCF)** is below 94%. Doing so will result in the craft accelerating to 0.4c instantly, liquefying the biological pilot due to inertial lag. The Inertial Nullifier (RMD-03 §2) cannot compensate for velocities exceeding 0.3c due to the breakdown of the graviton shielding at relativistic speeds.

GWA Parameter	Nominal Value	Minimum Safe	Maximum Rating	Notes
Graviton Flux Density	4.72 TeV	4.50 TeV	5.10 TeV	Must remain within ± 0.15 TeV of nominal
Pulse Duration (Maneuver)	0.04 s	0.01 s	0.12 s	See RMD-04 §1.1 for turn procedures
Pulse Frequency (Omni-Drift)	12.4 Hz	10.0 Hz	15.0 Hz	See RMD-04 §3 for calibration
Power Draw	847 MW	600 MW	1,200 MW	Exceeding 1,200 MW drains Casimir reserves
G-Force Compensation	40,000 G	1,000 G	65,000 G	Beyond 65,000 G, graviton coherence fails

1.4 Casimir Cavity Array Specifications

The Casimir Array consists of **2,048 individual Cavities** arranged in a spherical topology within the reactor housing. Each cavity operates at a plate separation of 10.0 nanometers $\pm$ 0.02 nm. The quantum vacuum fluctuations between the plates generate a net force that is amplified by the Element 115 decay products into usable graviton flux.

Cavity Parameter	Value	Tolerance	Measurement Method
Plate Separation	10.0 nm	$\pm$ 0.02 nm	Interferometric laser calibration
Plate Material	Degenerate Tantalum-Carbide	N/A	N/A
Operating Temperature	4.2 K	$\pm$ 0.1 K	Cryogenic thermocouple array
Quantum Efficiency	97.3%	$\pm$ 1.2%	Graviton flux measurement
Array Topology	Spherical (Icosahedral symmetry)	$\pm$ 0.001°	Structural interferometry
Active Cavities (Minimum)	1,800	N/A	Any fewer triggers emergency shutdown

2. STARTUP SEQUENCE (COLD START)

A Cold Start is defined as any startup from a fully powered-down state (core temperature > 50 K). Warm Starts (core temperature 4.2–10 K) follow a abbreviated sequence per §2.1. All steps require verbal confirmation from the Astrometrics Lead (RMD-02) and the Chief Medical Officer (RMD-03) before proceeding.

2.1 Cold Start Procedure

Step	Action	Verification	Responsible Party	RMD Reference
1	Verify MCF integrity	Must read >98.5% on all 12 sensors	Power Systems Tech	RMD-01 §1.1
2	Activate cryogenic cooling loop	Helium temperature < 2.1 K	Cryogenics Engineer	N/A
3	Verify Casimir Cavity gap alignment	All 2,048 cavities within ± 0.02 nm	Power Systems Tech	RMD-01 §1.4
4	Initiate Moscovium feed at 2.0 mg/hr	Feed rate stable for 60 seconds	Reactor Operator	RMD-01 §1.1
5	Monitor SNCL lattice integrity	Must read >99.0%	Power Systems Tech	RMD-01 §1.1
6	Wait for Decay Rate to stabilize	4.72 TeV $\pm$ 0.05 TeV for 120 seconds	Astrometrics Lead	RMD-02 §1
7	Verify GWA pre-charge status	All 4 projectors at standby	Flight Test Director	RMD-04 §1
8	Confirm Neural Shunt link	CSI sync coherence >98%	Chief Medical Officer	RMD-03 §1
9	Engage MCF full power	MCF >99.0% sustained	Power Systems Tech	RMD-01 §1.1
10	Increase Moscovium feed to 8.0 mg/hr	Decay rate stable at 4.72 TeV	Reactor Operator	RMD-01 §1.1
11	Verify tachyon flux levels	< 0.5 micro-Sieverts	Health Physics Tech	RMD-05 §1
12	Engage Casimir Array full power	2,048 cavities active, 97%+ efficiency	Power Systems Tech	RMD-01 §1.4
13	Increase Moscovium feed to 12.5 mg/hr (nominal)	Full graviton flux achieved	Reactor Operator	RMD-01 §1.1
14	Final systems check	All parameters nominal	All departments	All RMD docs
15	Flight clearance	Written authorization from Base Commander	Base Commander	RMD-05 §3

2.2 Warm Start Procedure

If core temperature is between 4.2 K and 10 K, Steps 1–3 of the Cold Start are skipped. Begin at Step 4 with a feed rate of 4.0 mg/hr and increase incrementally over 30 minutes.

2.3 Abort Conditions

IF any of the following conditions are met during startup, **THEN** immediately initiate Emergency Shutdown per §3:

- MCF drops below 94.0% at any point
- Decay rate exceeds 4.90 TeV or drops below 4.50 TeV
- Tachyon flux exceeds 2.0 micro-Sieverts
- Casimir Cavity active count drops below 1,800
- CSI sync coherence drops below 80% (see RMD-03 §1.1)
- Any single GWA projector fails to pre-charge within 300 seconds

3. EMERGENCY SHUTDOWN PROCEDURE

The Emergency Shutdown (designation: **COLD IRON**) is a controlled de-escalation of all reactor systems. COLD IRON is distinct from Protocol Omega (RMD-05 §3), which is a full scuttle. COLD IRON preserves the craft and all systems.

Step	Action	Time Required	Notes
1	Halt Moscovium feed immediately	0.0 s	Solenoid valve closes in 12 ms
2	Disengage GWA pre-charge	0.5 s	All 4 projectors to standby
3	Reduce MCF to 80%	2.0 s	Controlled field reduction
4	Vent Casimir Array cryogenic loop	5.0 s	Helium vented to external heat sink
5	Monitor Decay Rate descent	Ongoing	Must fall below 4.00 TeV within 60 s
6	Confirm SNCL lattice cooling	Ongoing	Lattice must not exceed 15 K during cooldown
7	Power down QEA navigation buffer	10.0 s	Per RMD-02 §3
8	Disconnect CSI neural link	15.0 s	Per RMD-03 §1 — pilot may experience brief nausea
9	Secure all tachyon siphons	20.0 s	Per RMD-05 §1 — prevents temporal bleed
10	Final power-down confirmation	60.0 s	All systems green, core temp < 10 K

IF COLD IRON fails to complete within 120 seconds, **THEN** escalate to Protocol Omega (RMD-05 §3). **DO NOT** attempt a second COLD IRON sequence — repeated thermal cycling degrades the SNCL lattice beyond recovery.

4. CASIMIR CAVITY FLUCTUATION TROUBLESHOOTING MATRIX

The following matrix addresses known fluctuation modes in the Casimir Cavity Array. All troubleshooting must be performed with the GWA fully disengaged. If GWA engagement is detected during cavity maintenance, the array will automatically lock out and the reactor will enter COLD IRON per §3.

Symptom	Likely Cause	Diagnostic Check	Corrective Action	Safety Precaution	RMD Reference
Oscillating graviton flux (± 0.3 TeV)	Cavity gap drift in sectors 7–12	Interferometric scan of all 2,048 cavities	Recalibrate affected cavities using the He-3 micro-shunt protocol	Verify MCF >98% before recalibration	RMD-01 §1.4
Localized temperature spike (>6 K in one sector)	Cryogenic micro-leak in helium loop	Pressure test individual helium feed lines	Seal leak with Class-VII epoxy; replace feed line if >0.5 mm breach	Evacuate sector; monitor tachyon flux	RMD-05 §1
Phase decoherence across >100 cavities	QEA timing desync (RMD-02 §3)	Cross-reference QEA clock with Casimir timing bus	Resync QEA master clock to Casimir reference oscillator	Do not exceed 2 resync attempts per hour	RMD-02 §3
Sustained low-frequency hum (12–15 Hz)	Mechanical resonance in cavity mounting structure	Structural vibration analysis	Adjust cavity mounting dampeners; tighten torque bolts to 42 N·m	Ensure all personnel clear of reactor housing	N/A
Intermittent power drops (<600 MW for <0.5 s)	MC-299 feed rate fluctuation	Inspect solenoid valve and feed line for obstruction	Purge feed line with inert gas; replace solenoid valve if >3 cycles of intermittent failure	Initiate COLD IRON if >5 power drops in 10 minutes	RMD-01 §1.1
Tachyon backflow into cavity array	Siphon failure (RMD-05 §1)	Measure tachyon flux at cavity inlet	Repair or replace tachyon siphon; flush cavity array with anti-tachyon field	Evacuate all personnel within 50 m radius	RMD-05 §1, §3
Cavities collapsing to <8.0 nm gap	SNCL lattice degradation	Full SNCL integrity scan	Initiate controlled SC-7 lattice reformation (requires 48-hour reactor downtime)	Begin COLD IRON sequence; no restart without Director approval	RMD-01 §1.1

Symptom	Likely Cause	Diagnostic Check	Corrective Action	Safety Precaution	RMD Reference
Cavities expanding to >12.0 nm gap	Quantum vacuum pressure anomaly	Cross-check with gravitational wave background	Adjust cavity electrostatic tensioning system (ETS) by +0.3 V increments	Do not exceed 13.0 nm — false vacuum risk	RMD-01 §1.1
Random cavity activation/deactivation	Electromagnetic interference from external source	Spectrum analysis of EM environment within 200 m	Activate EM hardening protocol; check for unauthorized RF transmissions	Report to Security per RMD-05 §4	N/A
Cavity efficiency drops below 90%	Degenerate Tantalum-Carbide plate oxidation	Microscopic inspection of plate surfaces	initiate plasma reconditioning cycle (requires Class-IV clean room)	Do not open reactor housing without MCF at 80%+	RMD-01 §1.4

DIRECTOR'S NOTE: *The cavity collapse scenario (row 7) was observed during the 2025-11-03 incident at Site-7. The 48-hour downtime requirement is non-negotiable. Dr. Vasquez's team requires full access during reformation. — Maj. Gen. Halloway*

5. POWER DISTRIBUTION HIERARCHY

The TARTARUS-1 power distribution follows a strict hierarchy. If power draw exceeds total Casimir Array output (nominal: 1,200 MW), the system will shed load in the following order:

Priority	System	Power Draw (MW)	Shed Threshold	Recovery Procedure
1 (Lowest)	Environmental Control & Life Support (ECLS)	45	Last to shed	Automatic restoration upon power availability
2	Sensor Arrays (Radar, LIDAR, Quantum)	82	Shed if power < 900 MW	Manual reactivation from pilot station
3	Hull Metamaterial Active Camouflage	120	Shed if power < 800 MW	Manual reactivation from pilot station
4	QEA Navigation Buffer	185	Shed if power < 700 MW	Requires QEA restart per RMD-02 §3
5	Inertial Nullifier	210	Shed if power < 600 MW	DANGER — See RMD-03 §2
6	CSI Neural Link	95	Shed if power < 500 MW	DANGER — See RMD-03 §1
7 (Highest)	GWA Propulsion	470	Never shed	N/A — Primary mission system

IF power drops below 500 MW and the CSI is shed, **THEN** the pilot loses all cognitive control of the craft. The QEA will enter autonomous navigation mode and attempt to return to the nearest friendly base. IF no friendly base is within Class II fold range, **THEN** the QEA will initiate Protocol Omega automatically.

6. MAINTENANCE SCHEDULES

System	Inspection Interval	Maximum Service Interval	Responsible Party	RMD Reference
Casimir Cavity Array	72 hours	168 hours (7 days)	Power Systems Tech	RMD-01 §1.4
SNCL Lattice	24 hours	72 hours	Reactor Operator	RMD-01 §1.1
MCF Sensors (12-unit array)	12 hours	48 hours	Power Systems Tech	RMD-01 §1.1
GWA Projectors (4 units)	Pre-flight only	50 flight hours	Flight Test Director	RMD-04 §1
Moscovium Feed System	Pre-flight only	100 mg throughput	Reactor Operator	RMD-01 §1.1
Tachyon Siphon Assembly	24 hours	72 hours	Health Physics Tech	RMD-05 §1
Cryogenic Helium Loop	168 hours	720 hours (30 days)	Cryogenics Engineer	RMD-01 §2

7. CROSS-REFERENCE INDEX

This document references the following sections in other RMD documents:

Reference	Document	Section	Topic
RMD-02 §1	Spatial Manifold Navigation	Non-Euclidean Routing	QEA graviton flux requirements
RMD-02 §3	Spatial Manifold Navigation	Navigation Interface	QEA timing bus synchronization
RMD-03 §1	Neural Sync & Inertial Damping	Cortical Shunt Interface	CSI sync coherence requirements
RMD-03 §2	Neural Sync & Inertial Damping	Inertial Nullifiers	Nullifier power requirements and G-force limits
RMD-04 §1	Atmospheric & Trans-Medium Maneuvers	Standard Operating Procedures	GWA pulse timing for right-angle turns
RMD-04 §3	Atmospheric & Trans-Medium Maneuvers	Omni-Drift	GWA pulse frequency for drift calibration
RMD-05 §1	Emergency Containment & Scuttle	Tachyon Radiation	Tachyon exposure limits and siphon protocols
RMD-05 §2	Emergency Containment & Scuttle	MCF Failure	Black hole event at <50% MCF
RMD-05 §3	Emergency Containment & Scuttle	Protocol Omega	Full scuttle procedure

APPENDIX A — ACRONYMS AND GLOSSARY

Acronym	Full Term	Definition
Bh	Bohrium	Synthetic element (Z=107) produced during Mc-299 decay chain
COLD IRON	Controlled Operations for Low-power Deactivation	Emergency shutdown procedure that preserves the craft
CSI	Cortical Shunt Interface	Surgical brainstem implant enabling pilot-craft cognitive link
Db	Dubnium	Synthetic element (Z=105) produced during Mc-299 decay chain
DUMB	Deep Underground Military Base	Classification for subterranean facilities housing TARTARUS-1
ECLS	Environmental Control & Life Support	Cabin atmosphere, temperature, and pressure regulation
ETS	Electrostatic Tensioning System	Mechanism maintaining Casimir Cavity plate separation
GWA	Gravity Wave Amplifier	Kerr-Newman Metric Projector converting ZPE to directional thrust
He-3	Helium-3	Isotope used in micro-shunt cavity recalibration protocol
Mc	Moscovium	Element 115 — primary fuel catalyst for ZPE generation
MCF	Magnetic Confinement Field	12-sensor array confining Element 115 within the decay chamber
Nb	Niobium	Element (Z=41) produced during Mc-299 decay chain; secondary tachyon emitter
NHI	Non-Human Intelligence	Classification for the designers/operators of TARTARUS-1
Omega	Protocol Omega	Full scuttle procedure — fatal to pilot, destroys all materiel
QEA	Quantum-Entangled Astrometrics	Navigation computer calculating spatial manifold fold metrics
RCS	Radar Cross-Section	Measure of detectability on adversary radar systems
RMD	Recovered Materiel Division	Vanguard Aerospace division responsible for TARTARUS-1 operations
SCI	Sensitive Compartmented Information	Classification above Top Secret; compartmented access required
SNCL	Strong Nuclear Confinement Lattice	Crystalline matrix stabilizing Mc-299 half-life

Acronym	Full Term	Definition
Sr	Strontium	Element (Z=38) produced during Mc-299 decay chain
TeV	Tera-electronvolt	Unit of energy (10^{12} eV) used to measure graviton flux
ZPE	Zero-Point Energy	Quantum vacuum energy harvested by the Casimir Cavity Array

APPENDIX B — ERROR CODES

Code	System	Description	Severity	Required Action
PWR-001	MCF	Confinement field below 94.0%	CRITICAL	Initiate COLD IRON (§3)
PWR-002	MCF	Single sensor failure	WARNING	Replace sensor within 24 hours
PWR-003	MCF	Triple sensor failure	CRITICAL	Initiate COLD IRON; ground all flights
PWR-010	Casimir	Cavity gap deviation >0.05 nm	WARNING	Recalibrate per §4 matrix
PWR-011	Casimir	Cavity gap deviation >0.10 nm	CRITICAL	Initiate COLD IRON
PWR-012	Casimir	Active cavities <1,800	CRITICAL	Initiate COLD IRON; 48-hour downtime
PWR-020	Decay	Rate <4.50 TeV	CRITICAL	Increase Mc-299 feed; if no response, COLD IRON
PWR-021	Decay	Rate >5.00 TeV	CATASTROPHIC	Initiate Protocol Omega (RMD-05 §3)
PWR-030	Feed	Mc-299 rate <10.0 mg/hr	CRITICAL	Purge feed line; if blockage persists, COLD IRON
PWR-031	Feed	Mc-299 feed interruption >5 s	CRITICAL	Initiate COLD IRON
PWR-040	GWA	Projector pre-charge failure	WARNING	Retry pre-charge; if >2 failures, ground flight
PWR-041	GWA	Projector output >1,200 MW	CRITICAL	Reduce thrust; if sustained, COLD IRON
PWR-050	Tachyon	Flux >2.0 micro-Sieverts	WARNING	Evacuate sector; inspect siphon
PWR-051	Tachyon	Flux >8.0 micro-Sieverts	CATASTROPHIC	Initiate Protocol Omega (RMD-05 §3)
PWR-060	SNCL	Lattice integrity <98.0%	WARNING	Schedule maintenance within 48 hours
PWR-061	SNCL	Lattice integrity <95.0%	CRITICAL	Initiate COLD IRON; 72-hour SC-7 reformation
PWR-070	Power	Total output <500 MW	CRITICAL	System sheds load per §5 hierarchy
PWR-071	Power	Total output <200 MW	CATASTROPHIC	QEA auto-initiates Protocol Omega

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